

L^AT_EX for Elegant Humanities Handouts in Conferences and Classrooms

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Abstract

L^AT_EX has been popular throughout the mathematical and physical sciences for decades. L^AT_EX’s utility for scholars in humanities disciplines, however, tends to be overlooked. In this tutorial, I introduce the use of L^AT_EX for a common task in humanities scholarship, namely, typesetting materials with unusual line length and other organizational needs, such as poetry and translations. The specific example is preparing a handout for a historical conference or class presentation with excerpts of ancient Greek and parallel English translations. Additionally, I provide general recommendations for learning and employing L^AT_EX across the academic disciplines of the humanities. I aim to inspire colleagues in religious, historical, philosophical, and related areas of study to think of L^AT_EX as having a wide range of professional applications even inside these scholarly fields in which the use of L^AT_EX is not widespread.

1 L^AT_EX for the humanities

The world’s best established software system for typesetting and document formatting, L^AT_EX has been popular throughout mathematical and physical sciences since before I was born. The utility of L^AT_EX for humanities disciplines, however, long has been overlooked. The barrier to entry of learning a language for computer typesetting often means that persons outside disciplines do not bother. Yet, L^AT_EX can be used for elegant typesetting in most instances in which a scholar of humanities needs to present materials, especially when the presentation matters per se. Whether one works in history, philosophy, literature, or some other discipline in which L^AT_EX is not usually employed, L^AT_EX is worth taking the time to learn.

Granted, in disciplines outside the mathematical and physical sciences, such as my own area of historical theology (technically, divinities rather than humanities), editors expect us to prepare our materials for publications with bug-for-bug compatibility with Microsoft Word for back-and-forth review and markup. In practice, this expectation requires us to use word processors, such as LibreOffice Writer (among other free and open source options). A word processor presents material in a What You See Is What You Get (WYSIWYG, “whizzy-wig”) style. While one writes, one simultaneously thinks about and tends to matters of formatting.

In a typesetting language, such as L^AT_EX (or ConT_EXt or Typst, either of which a scholar also might choose for various reasons), one is only concerned about the structure of what one is writing. One declares that specific characters will be in certain categories of writing ‘things’ (heading, paragraph, footnote, accent type, typeface, and so on), and one allows the program to compile one’s writing as ‘code’ into a tidy final document — every time. Preparing a *curriculum vitae*, conference handout, or other document with typesetting requirements beyond what are straightforward to achieve in a word processor becomes more about the writing itself with less worry about the formatting.

To learn L^AT_EX as a scholar outside the disciplines in which L^AT_EX is the norm, I recommend working through one of two community-built, libre guidebooks. The first, which I used to learn L^AT_EX, is *The Not So Short Introduction to L^AT_EX* (github.com/oetiker/lshort). The second, which a colleague of mine recommends, is *Modern L^AT_EX* (github.com/mrkline/modern-latex). Either guidebook may be used to begin to learn to typeset cover letters, ancient languages, and other materials that are relevant across the humanities — without the front-loading of mathematics and physical sciences that one finds in most other introductions.

To show how thinking about digital typesetting in terms of document structure rather than final format can be of practical use in the humanities, here I present a tutorial in which I introduce the use of L^AT_EX for a common task in humanities scholarship, namely, typesetting texts with unusual line length and other organizational needs. The case study is preparing a handout for a historical conference or class presentation with perfectly organized excerpts of a Byzantine Greek text and an original English translation in parallel. The principles that I outline will be applicable to any presentation of parallel text and, more generally, all types of handouts that are common in humanities conferences and classrooms.

2 Simplest side-by-side text in L^AT_EX:

`parallel`

There are many ways to set parallel text in L^AT_EX. The simplest is to use the community package called `parallel`, which is included in T_EX Live, MiK_T_EX, and other L^AT_EX distributions. For full documentation, visit ctan.org/pkg/parallel.

To start using `parallel` in any L^AT_EX document, include the following in the preamble:

```
\usepackage{parallel}
```

For example, here is an abbreviation of the preamble to the Beamer file for my talk at the 2026

conference of the T_EX Users Group (github.com/historical-theology/latex-talks/) with only the most important entries, including the declaration of the package `parallel`:

```
\documentclass[aspectratio=169]{beamer}
\usetheme{Goettingen}
```

```
\usepackage{parallel}
```

```
\title{\LaTeX\ for Elegant
  Humanities Handouts \\\
  in Conferences and Classrooms }
\author{Corey Stephan, Ph.D.}
```

```
\begin{document}
```

This is the basic format for parallel text set with the package `parallel`:

```
\begin{Parallel}{<L space>}{<R space>}
\ParallelLText{<left text row 1>}
\ParallelRText{<right text row 1>}
\ParallelPar
\ParallelLText{<left text row 2>}
[and so on]
\end{Parallel}
```

I (a theologian) might typeset the opening words of the Christian prayer *Ave Maria* in their traditional Latin with parallel English like so:

```
\begin{Parallel}{0em}{0em}
\ParallelLText{Ave Maria,}
\ParallelRText{Hail Mary,}
\ParallelPar
\ParallelLText{gratia plena \ldots}
\ParallelRText{full of grace \ldots}
\ParallelPar
\end{Parallel}
```

Here is the result:

Ave Maria,	Hail Mary,
gratia plena ...	full of grace ...

`parallel` makes it easy to set short sections of text side-by-side in L^AT_EX.

3 Advanced parallel text in L^AT_EX: `reledpar` and `reledmac`

For more advanced setting of parallel text in L^AT_EX, especially long-form (e.g. a critical edition with accompanying translation), one ought to learn the two interconnected and frequently updated community packages `reledpar` (for parallel typesetting specifically) and `reledmac` (for typesetting academic editions in general), which are bundled with the major L^AT_EX distributions.

`reledpar` and `reledmac` are purpose-built for lengthy side-by-side texts, especially in academic articles, books, and critical editions. These two packages are meant to replace older methods of academic parallel text preparation in L^AT_EX that one might find in Web searches. For full documentation, visit ctan.org/pkg/reledpar and ctan.org/pkg/reledmac.

4 Setting non-Latin characters in L^AT_EX: `babel` and `polyglossia`

The most important L^AT_EX packages for languages with non-Latin alphabets are `babel` and `polyglossia`, which are included in the major L^AT_EX distributions. Although it is possible to convert a project from one of these metapackages to the other, a scholar ought to invest the time to choose carefully between either of these two at the start of a grand project.

In *The L^AT_EX Companion: 3rd Edition* (Addison-Wesley, 2023), Frank Mittelbach and Ulrike Fischer suggest that despite the fact that `babel` is older than `polyglossia`, `babel` generally is preferable: “Unless you are in a special situation where a language or script is notably better supported by `polyglossia`, the recommendation is to use `babel` with all engines” (entry 13.7.2, “Polyglossia”). Since `babel` technically supports the same languages as `polyglossia`, a situation in which one would need `polyglossia` over `babel` is rare. Yet, a scholar working with an ancient language might face a situation in which, say, `polyglossia` actually supports that language better than `babel`. To make an informed choice against a given academic undertaking’s particular requirements, one should explore each package’s documentation at ctan.org/pkg/babel and ctan.org/pkg/polyglossia.

In (old) Web forum discussions, one might read that `babel` will work better with the L^AT_EX engine LuaL^AT_EX, whereas `polyglossia` will work better with XeL^AT_EX. As of the time of writing this essay, however, one may choose either of the two mainline L^AT_EX engines that are built with full Unicode support, XeL^AT_EX or LuaL^AT_EX, for either `babel` or `polyglossia`. Results might vary depending on the specifics of a given project.

To learn `babel`, I recommend Chapter 13: “Localizing documents” in *The L^AT_EX Companion: 3rd Edition*. Here, Mittelbach and Fischer succinctly introduce localization in L^AT_EX via `babel` in a practical, didactic style. Of course, the package’s official documentation in CTAN also ought to be consulted for assistance with complex arrangements.

To learn `polyglossia`, I recommend the package’s documentation in CTAN.

5 Starting a project with polyglossia

Here, I will model `polyglossia`, which is what I happen to be familiar with using for polytonic (ancient and Byzantine) Greek.

First, include the package in the preamble:

```
\usepackage{polyglossia}
```

Then, after the beginning of the document but before actual type, declare the languages and their variants. For example, here is this .tex file's declarations:

```
\begin{document}
\setdefaultlanguage{english}
\setotherlanguage[variant=ancient]{greek}
```

One is able to use one of the 'other' declared languages (in this case, ancient Greek) by toggling it when needed.

A declared foreign or ancient language may be toggled for a long excerpt like so:

```
\begin{greek}
<greek text>
...
\end{greek}
```

For example, this is how one could set the famous opening verse of John's Gospel:

```
\begin{greek}
ἐν ἀρχῇ ἦν ὁ λόγος, \
καὶ ὁ λόγος ἦν πρὸς τὸν θεόν, \
καὶ θεὸς ἦν ὁ λόγος. \
\end{greek}
```

Here is the clean result:

ἐν ἀρχῇ ἦν ὁ λόγος,
καὶ ὁ λόγος ἦν πρὸς τὸν θεόν,
καὶ θεὸς ἦν ὁ λόγος.

For another example, here are the first two lines of Homer's *The Iliad*:

```
\begin{greek}
μῆνιν ἄειδε θεὰ Πηληϊάδεω Ἀχιλῆος \
οὐλομένην, ἣ μυρὶ 'Ἀχαιοῖς ἄλγε' ἔθηκε,
```

The result is equally clean to that of John 1:1, even for these particular words of Homer with their more complicated accent and breathing marks:

μῆνιν ἄειδε θεὰ Πηληϊάδεω Ἀχιλῆος
οὐλομένην, ἣ μυρὶ 'Ἀχαιοῖς ἄλγε' ἔθηκε,

A declared language may be toggled for a single word or line (rather than a series of verses or lines) in this way:

```
\text<language>{<text in language>}
```

This mid-line toggling style be used to insert a key-word in that declared language, e.g. ὁ λόγος (produced with `\textgreek{ὁ λόγος}`), inside running prose in another language, such as the document's primary one.

6 Combining parallel and polyglossia

The simplest way of setting multilingual text side-by-side is with `parallel`, as was shown in section (2). Since `parallel` can be tedious to organize for more than a stanza/paragraph or two, I remind the reader of `reledpar` and `reledmac` for long and/or complicated parallel text displays, as I suggested in section (3).

Here is an example of `parallel` and `polyglossia` working in combination to display my own translation from the opening chapter of my recently published historical monograph *Maximus the Confessor's Thomistic Legacy*:

```
\begin{Parallel}{0em}{0em}
\textgreek{\ParallelText
{Νῦν συγχορεύεις ἀληθῶς}}
\ParallelRText
{Now, truly, you celebrate together}
\ParallelPar
[and so on]
\end{Parallel}
```

For a conference or classroom handout, the result might look like Figure 1: "Canon in Honor of Thomas Aquinas," Ode 3, which is shown at full page width on the next page.¹

7 Other L^AT_EX tools for humanities handouts

There are many L^AT_EX tools that can assist work across the humanities disciplines. Here is a succinct, curated list of the most broadly applicable, especially those that one might wish to use to assist with preparing a handout for a conference presentation or university class period:

- Either BIBL^AT_EX or the older but still useful BIBT_EX offers a comprehensive citation and bibliography management system that is capable of being interlinked with a variety of citation managers — notably, for academic research, Zotero and JabRef.

¹ The Greek text in Figure 1 is from Raffaele Cantarella, ed., "Canone greco inedito di Giuseppe vescovo di Methone in onore di San Tommaso d'Aquino," *Archivum Fratrum Praedicatorum* 12 (1934): 145–85, at 155–58. This parallel Greek text with my Latin translation duplicates Corey J. Stephan, *Maximus the Confessor's Thomistic Legacy*, *Medieval Law and Theology* 10 / *Studies and Texts* 244 (Toronto: Pontifical Institute of Mediaeval Studies, 2026), 6.

Νῦν συγχορεύεις ἀληθῶς
 σὺν τοῖς λοιποῖς θεολόγοις
 καὶ τοῖς ἄλλοις θεολήπτοις ἀνδράσι,
 περὶ τὸν θρόνον Χριστοῦ
 τὸν ἱερὸν ἐξῆδοντες
 καὶ τρισάγιον ὕμνον
 σὺν ταῖς ἀγίαις δυνάμεσιν.

Ἐνθα πατέρων ὁ χορὸς
 καὶ ἱερῶν θεολόγων
 ὡς ὑψίνους καὶ λεπτὸς θεόλογος
 κατεσκήνωσας, Θωμᾶ,
 σὺν τούτοις ἀυλιζόμενος,
 θεολογῶν ἀπαύστως
 τριάδα, πάτερ, τὴν ἄκτιστον.

Πῦρ ἀνεδείχθης φλογερὸν
 τὰς φρυγανώδεις αἱρέσεις
 τῇ ἰσχύϊ τῆς σοφίας σου, πάτερ,
 κατακαύσας, καὶ πιστῶν
 καταφαιδρύνας, ἅγιε,
 τὰς διανοίας ὅθεν
 σὲ συνελθόντες γεραίρομεν.

Now, truly, you celebrate together
 with the rest of the theologians
 and the other divinely inspired men
 around the throne of Christ,
 singing the sacred
 and Thrice-Holy Hymn
 with the holy powers.

Where there is the chorus of Fathers
 and holy theologians,
 as a high-minded and subtle theologian
 you do abide, Thomas,
 dwelling with them,
 unceasingly theologizing,
 O Father, about the Uncreated Trinity.

A burning fire you have shown yourself to be
 against the firewood-like heresies
 by the might of your wisdom, O Father,
 you have burned them down, and
 you have restored completely, O Saint,
 the thoughts of the faithful, whence
 we, all joined together, do venerate you.

Figure 1: “Canon in Honor of Thomas Aquinas,” Ode 3

BIB $\LaTeX$ and BIB $\TeX$ are both built into $\TeX$ Live and other $\LaTeX$ distributions.

For work in humanities disciplines, BIB- $\LaTeX$ has the advantage over BIB $\TeX$ of full Unicode (UTF-8) support. Yet, if one does not need full UTF-8, one ought to consider either option.

- The free and open source (GPL-2) graphical software application **TeXstudio** (texstudio.org) offers a $\LaTeX$ -specific writing environment with comfortable features like intuitive autocompletion of $\LaTeX$ commands.

$\LaTeX$ may be written in any plain text editor, but a designated environment improves efficiency and helps avoid code errors.

I specifically mention **TeXstudio** because it is what I use for my own work and (on a practical note) it is included in the official software repositories of most libre Unix-like operating systems (GNU/Linux and BSD).

- By running search queries in CTAN, one may find community-built packages for many individual languages that are of relevance for historical or other academic research, e.g. **aramaic-serto** for the ancient Middle-Eastern language Syriac (Aramaic).

- **microtype** offers many small adjustments to type, such as improved automatic justification and hyphenation.

This TUGboat article provides a working example of the **microtype** package in-use.

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